

Distress Disclosure across Social Media Platforms during the COVID-19 Pandemic: Untangling the Effects of Platforms, Affordances, and Audiences

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ABSTRACT

Understanding how and why people share negative emotions and thoughts on social media has received much scholarly attention. Scholars have identified a variety of factors that affect disclosure behavior, but as platforms offer a wider range of affordances that enable more diverse user behaviors and nuanced audience segmentation, these influencing factors are increasingly intertwined. However, little is known about the interrelatedness of platform, affordance, and audience. Drawing on survey data of 470 American adults during the COVID-19 pandemic, this study examines the interplay and relative strength of the factors influencing distress disclosure on social media. We introduce the concept of *social media disclosure ecology* as an analytical lens to understand online disclosure. The results suggest that perceived affordances (i.e., anonymity, persistence, visibility control) and relational closeness to audience separately and interactively predict the depth of distress disclosure, which in turn affects satisfaction with disclosure. This study contributes to the literature on online-disclosure and privacy, while providing implications for the design of social media to better support people in distress.

CCS CONCEPTS

• **Human-centered computing**; • **Social media; Human-centered computing**; • **Empirical studies in collaborative and social computing**;

KEYWORDS

Social media, self-disclosure, stress, mental health, affordances, social networks

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1 INTRODUCTION

With the rapid spread of the coronavirus disease 2019 (COVID-19), individuals in the US have lived through multiple overlapping crises: a global pandemic, economic downturn, racist violence, and massive social justice protests. These stressors have taken a toll on individuals' mental health. According to a recent national survey [102], one-third of Americans (33%) have experienced high levels of psychological distress during the coronavirus outbreak. Substance use and suicidal ideation have also increased [34]. The heightened stress and anxiety, compounded by lockdowns and social distancing directives, have induced a startling surge in social media usage [99, 100]. For many people, social media have become a lifeline in seeking social connections and support [18, 92], which is evidenced by the large amount of disclosures of distress related to COVID-19 on social media [18, 108]. In response to the increasing number of disclosures of distress, Facebook has launched new "care" emoji reactions to help users support one another [85].

The prevalence of distress disclosure on social media is not surprising. Ample empirical evidence suggests that negative emotions can fuel verbal expression and social interactions, which may in turn help alleviate these emotions [30, 32, 104, 111, 131]. Scholars in human-computer interaction (HCI) and related fields have long been interested in understanding people's motivations, experiences, and consequences of disclosing their negative experiences and/or emotions online, such as depression [35, 52, 132], eating disorders [26, 97], pregnancy loss [6], and sexual abuse [7, 58]. Of particular interest to scholars are the factors that influence people's disclosure decisions and behaviors. While traditional theories of self-disclosure have primarily focused on personal goals and motivations for self-disclosure [38, 96], recent empirical evidence emphasizes the pivotal role of contextual and technological factors in affecting online disclosures [5, 6, 13, 16, 89].

Much of the literature on distress disclosure on social media has focused on a single platform [6, 9, 36], yet a growing body of work concerns the constellation of the platforms utilized by users [40, 106, 107, 134] and the varied disclosure behaviors across platforms [29, 57, 119]. Although the norms and meanings surrounding different platforms vary [19], user behavior has become increasingly diversified on individual platforms because of a wider array of features and affordances these platforms offer [12, 55]. For instance, Instagram offers a range of features that enable the sharing of persistent pictures, ephemeral posts (i.e., Stories), private messaging, and short videos (i.e., Reels). Therefore, users may exhibit distinctive behaviors on the same platform depending on the channel or feature they use and its audience. Another layer of this complexity is the widespread practice of maintaining multiple accounts on

a particular platform to manage different identities and distinct groups of audiences [7, 43, 64, 105].

The ever-evolving social media ecology has engendered a dynamic interplay of forces that shape online disclosure, or what we term *social media disclosure ecology*. While these forces can be conceptualized in different ways [6, 89, 134], they all point to the importance of social contexts and situational factors that are constantly in flux. The theory of situational privacy and self-disclosure differentiates situational factors (e.g., audience, emotions, and artifacts) and non-situational factors (e.g., demographics and personality traits) in affecting self-disclosure, suggesting that situational factors are the strongest predictors of disclosure behavior [89]. Moreover, qualitative research on HCI has shed light on multilevel factors that influence individuals' decisions about stigmatized disclosure [5, 6]. While these studies provide an in-depth, granular understanding of individuals' experiences and decision-making processes of distress disclosure, the relative importance of the influencing factors remains unclear. Because a variety of factors may directly and interactively affect human behavior, it is important to examine their unique influences and interactions.

In this study, we employ an ecological approach to examining the interplay of situational and non-situational factors in predicting distress disclosure on social media and its outcomes. Based on online surveys of a nationally representative sample of 470 American adults, we examine how situational factors (i.e., platforms, affordances, audience, and stressors) and non-situational factors (i.e., demographics, privacy concerns, and predisposition towards distress disclosure) have been associated with the depth of distress disclosure across 17 social media platforms during the COVID-19 pandemic, as well as the interactions between these factors. In addition, we examine the direct and indirect effects of these factors on satisfaction after distress disclosure.

This study contributes to the literature on self-disclosure and privacy in three ways. First, this study builds on and extends the theory of situational privacy and self-disclosure by integrating an affordance approach that transcends platform boundaries while also accounting for the effects of platforms. Second, it complements the existing HCI research on distress disclosure by uncovering the relative importance and interplay of the factors influencing disclosure behavior and outcomes. The improved understanding of when, why, and how people share personal information, especially in times of crisis, could inform interventions. Third, our findings shed light on the direct and indirect effects of perceived affordances on disclosure satisfaction, which provide implications for the design of sociotechnical systems to better support people in distress.

In this study, we adopt a broad definition of social media that includes social network sites (SNS), instant messaging (IM), and video chat platforms [55]. Previous research on distress disclosure has focused on SNSs [7, 35, 36], which tend to be more public than IM and video chat platforms. However, we argue that the boundary between public and private social media has become increasingly porous because of overlapping features and functionalities. For example, both Facebook and WhatsApp allow for private messaging and public status updates, which makes it difficult to draw a clear line between SNSs and IM. In this study, by taking a broad approach to social media, we were able to capture users' social media disclosure ecology to the fullest extent.

2 RELATED WORK

2.1 Distress Disclosure in Social Media Ecology

Self-disclosure is an integral part of interpersonal relationships. The process of revealing personal information to others is a crucial building block of intimacy [3, 74]. Self-disclosure includes revelations of personal experiences, thoughts, and feelings, which vary in terms of breath, depth, and duration [3, 126]. One type of self-disclosure that has received much scholarly attention is distress disclosure, which refers to the sharing of negative thoughts and feelings with others [30]. Distress disclosure has often puzzled researchers largely because of its paradoxical outcomes. On one hand, distress disclosure may alleviate stress and enhance coping by facilitating catharsis [98] and gaining social support from others [30, 75, 131]. On the other hand, distress disclosure carries inherent risks, such as social rejection and judgment, and excessive disclosure may diminish social support [6, 63, 74, 131], which leads to the dilemma of distress disclosure [30].

The dilemma of distress disclosure is amplified by social media, which enables one's messages to reach a larger and wider audience compared with offline situations [13, 47, 122]. In SNS communication, context collapse and positivity bias create additional challenges to sharing negative aspects of the self. However, researchers have observed a large number of disclosures of negative and stigmatized experiences and/or emotions on social media, such as depression [35, 52, 132], eating disorders [26, 97], pregnancy loss [6, 8], sexual abuse [7, 58], and LGBTQ identity [63, 64]. A growing body of research on HCI and communication has examined the motives, processes, and consequences of distress disclosure on social media [4, 9, 15, 40]. Computational scholars have also endeavored to detect and predict distress, mental illness, and suicidal ideation using large-scale digital trace data on social media [26, 37, 72, 78], which has been translated into real-world applications to detect at-risk individuals [124].

Because of the tradeoff in distress disclosure, much attention has been paid to factors that influence an individual's disclosure behavior. Traditional theories of self-disclosure posit that individuals pursue specific goals in self-disclosure and that different goals shape disclosure characteristics, such as breadth and depth [38, 96]. Disclosure goals vary depending on situational cues (e.g., social context) and individual differences (e.g., personal goals) [96]. Extending the theory to online settings, Bazarova and Choi [13] have proposed a functional model of self-disclosure on SNSs, suggesting that social media affordances activate disclosure goals, which then shape disclosure intimacy. Regarding stigmatized and emotionally difficult experiences, researchers have identified six factors that affect individuals' disclosure decisions, including the self, audience, network, platforms, society, and temporality, which are summarized in the disclosure decision-making (DDM) framework [6]. This framework has been employed to examine online disclosures of stigmatized experiences, such as pregnancy loss on Facebook [5, 8] and #metoo on Twitter [58], although the relative importance of the factors that influence such disclosures is unclear. Also missing in this framework are individual characteristics, such as sociodemographic factors, digital literacy, and privacy concerns, which may influence self-disclosure decisions and characteristics [42, 65, 133].

Complementing the DDM framework, the theory of situational privacy and self-disclosure [89] categorizes the antecedents of self-disclosure into situational and non-situational factors. Situational factors vary across situations and contexts, including personal factors (e.g., goals and emotions), interpersonal factors (e.g., audience), and external factors (e.g., objects and artifacts). Non-situational factors are usually stable and affect behavior in most situations, such as demographics and personality traits. Using a combination of smartphone tracking and experience sampling methods, Masur [89] found that the depth of self-disclosure was predicted mainly by situational factors, such as audience size and relational closeness, whereas non-situational personal factors were not significantly related to disclosure depth. Although it provides a comprehensive analysis of privacy and self-disclosure, the theory, as the author argued, may benefit from integrating the affordance approach to capture users' perceptions of the material features of the technology [89].

This study aims to expand the theory of situational privacy and self-disclosure by incorporating perceived affordances in the model. Specifically, we focus on situational factors, including personal factors (i.e., stressors), interpersonal factors (i.e., audience), and external factors (i.e., platforms and affordances), and non-situational factors, including demographics, privacy concerns, and propensity for distress disclosure. In the following sections, we discuss the factors included in our study and our hypotheses.

2.2 Social Media Platforms and Affordances

While earlier studies on distress disclosure have focused mainly on a single platform [6, 7, 35, 37], mounting evidence suggests that people incorporate multiple social media platforms into their communication practices to fulfill diverse goals and needs [19, 40, 105, 115, 134]. A plethora of social media research has found notable variations in user behavior across social media platforms in various contexts, such as protest participation [119], online disclosure [29, 105], self-presentation [40, 120], and news consumption [112, 129]. Individuals may use different platforms to pursue various goals [2]. For example, previous studies found that participants sought connectedness and intimacy on WhatsApp, and they sought self-esteem and popularity on Facebook [77]. Platforms are also associated with users' moods and feelings [11, 103]. For instance, Snapchat use predicted more positive moods compared with Facebook and Instagram interactions [11]. Moreover, previous research showed cross-platform differences in communicative outcomes [57, 68]. Users sought and received different dimensions of social support depending on the platform [68].

Despite these cross-platform differences, user behavior has become increasingly diversified on individual platforms because of a wider range of affordances [12, 55]. For example, because Facebook bundles together a range of public and private channels (e.g., Messenger, status updates, and wall posts), distinct usage patterns and user behaviors may emerge on this platform [13, 14]. Similarly, the ephemeral element that originated in Snapchat has been widely adopted by other social media platforms, such as Instagram Stories. Adding another layer of complexity, scholars have found the pervasive practice of maintaining multiple accounts on the same platform, especially regarding negative or stigmatized disclosures [6, 64, 105]. This finding suggests that between-platform differences in users' distress disclosures may not be salient. However, because of scarce evidence on the influence of platforms on distress disclosure, we ask the following:

RQ1: To what extent do specific platforms predict the depth of distress disclosure?

One approach that could transcend platform boundaries and still account for platform differences is the affordance perspective. Despite the many existing definitions of affordances [59, 95], researchers have increasingly recognized them as involving a relational concept that captures the interplay between an artifact and its user [51, 56, 116]. In other words, affordances emerge from a user's interactions with the materiality of an artifact [116]. The central tenet of affordances is the focus on users' perceptions of material features [95], which may influence how users interact with the system [39, 51]. By focusing on high-level characteristics across platforms, the affordance approach could help scholars generate findings that are generalizable beyond specific platforms, thus enabling a more flexible and durable theorization [55]. Although a wide array of affordances of social media have been identified in the literature [39, 56, 116], in the present study, we focus on three affordances—anonymity, persistence, and visibility control—that have been shown to be the most relevant in the context of distress disclosure [7, 105, 132].

Anonymity. The sensitive nature of distress disclosure often drives people to use anonymous platforms. Anonymity refers to the degree to which a user's real name or true identity can be concealed in a channel [81]. Anonymous sites grant users more flexibility in the identities they present [118]. Anonymity often leads to online disinhibition, in which people engage in less socially normative behavior, such as discussions of taboo topics [17, 81]. Users also tend to be more comfortable in making revelations about sensitive or stigmatized topics online when they are anonymous. For instance, studies have found that people tend to use "throwaway" accounts on Reddit to disclose mental health issues [7]. Given the sensitivity and vulnerability of distress disclosure, it is reasonable to assume that users may engage in more distress disclosures in sociotechnical systems that they perceive as affording greater anonymity.

Persistence. Persistence refers to the extent to which content is viewable after posting [116]. Social media that afford content persistence allow users to retrieve messages long after the original transmission. The opposite of persistence is ephemerality, which is characteristic of Snapchat or Instagram Stories, in which messages disappear after a certain amount of time. Studies have found that Snapchat was mainly used for entertainment purposes [2], and Snapchat interactions were associated with a more positive mood compared with other social media platforms, such as Facebook and Instagram [11]. Moreover, Snapchat was found to be used to share mundane experiences with close ties, and its users reported reduced self-presentation concerns [128]. In addition, the persistence of messages ensures that others may view or retrieve support-seeking content after a certain period, which may help garner more support [12]. Thus, it is reasonable to believe that ephemeral social media are less likely to be used for distress disclosure.

Visibility control. Visibility control refers to the extent to which a platform affords users to determine which content is visible to whom [39]. The perception of visibility control may be

contingent on the accessibility and use of the platform's privacy settings. Different levels of visibility control can lead to various disclosure behaviors on social media [88]. For instance, platforms that enable less control over the information flow and audiences may cause increased privacy concerns [29], which could prevent users from disclosing private and intimate information [133]. In addition, disclosing inappropriate information to unintended audiences could result in regret, failed relationships, and even job loss [123]. These negative outcomes are more likely to be caused by distress disclosure, which carries inherent risks because of its vulnerability and sensitivity [30]. Therefore, users are unlikely to share distress if they are unable to select the target audience [110]. Conversely, when perceived visibility control is high, users may have an increased willingness to express negative emotions and distress. For instance, users were found to share more intense and negative emotions in private messages than on public channels (i.e., status updates and wall posts) on Facebook [14]. Based on these previous findings in the literature, we hypothesize the following relationship between perceived affordances and distress disclosure on social media platforms:

H1: Perceptions of a) anonymity, b) persistence, and c) visibility control are positively associated with the depth of distress disclosure on social media.

2.3 The Impact of Social Context: Audience and Stress

In addition to platforms and affordances, distress disclosure may also differ across audiences. Sociologists have long contended that individuals present themselves differently based on their audience [61]. Such situational awareness also manifests in online contexts, as people engage in varied self-presentation depending on an actual or imagined audience (i.e., those they perceive to be viewing their content) [83, 84]. Prior research has shown that users tended to disclose more intimate information online to audiences to whom they felt close [29, 122]. However, the relationship between online disclosure and audience size is less clear. Some scholars found that a larger online network was a predictor of less self-disclosure [22, 29], while others showed a positive association between network size and the amount of disclosure [60, 122]. Such discrepancies might be attributable to the content of disclosure, as people tend to share more positive events/emotions with a larger network and share negative emotions with fewer others. Because of the private and sensitive nature of distress disclosure, it is reasonable to posit that people disclose more to a smaller audience and to those who are closer to them:

H2: (a) Larger and (b) relationally closer audiences are positively associated with the depth of distress disclosure on social media.

Stress is another situational factor that may affect disclosure behavior. Previous studies showed that stress was positively related to the time spent using IM [82] and self-disclosure on Facebook [131]. Technology use in times of stress is often driven by goals of mood management and support seeking [82], which might have been exacerbated during the unsettling circumstances of COVID-19. In seeking to determine the relationship between stress and online disclosure, we focused on external stressors experienced during the pandemic. The psychological literature distinguishes between stress exposure (i.e., external stressors) and stress response (i.e., perceived stress) [67]. Stress exposure concerns the environmental challenges to which an individual is exposed, whereas stress response measures subjective responses to these environmental challenges [67]. Because stress response can be a product of either exogenous environmental factors or social media use, measuring stress response without accounting for stress exposure [82, 131] complicates the relationship between stress and online disclosure. In other words, perceived subjective feelings of stress can be both an antecedent and an outcome of online disclosure, especially because the expression of negative emotions is often accompanied by a temporary increase in those emotions [28, 30]. By focusing on external stressors rather than perceived stress, we were able to decouple stress from technology use and determine the role of situational factors on disclosure. We hypothesized that experiencing a larger number of stressors during COVID-19 predicts more distress disclosure:

H3: The number of stressors is positively associated with the depth of distress disclosure on social media.

2.4 The Interplay of Situational and Non-situational Factors

While we aimed to differentiate the unique influences of situational factors on distress disclosure, it is also possible that these factors interact with each other. Notably, it is likely that perceived affordances and audience interactively influence disclosure behavior. Prior research showed that platforms and audiences were often intertwined [40, 134]. For example, one study found that content accessibility moderated the relationship between audience size and the depth of self-disclosure [89]. The varied settings of platforms, such as privacy boundaries, may determine the nature of the audience, which influences the content shared on the platforms. For instance, Facebook's loose privacy boundary may invite a large and diverse audience with whom a user may share little intimate information. Conversely, individuals may share their innermost feelings with loved ones on platforms that afford high visibility control over content, such as WhatsApp.

Moreover, situational stress may also interact with the audience in predicting disclosure behavior. For example, users who are highly stressed may broadcast their distress to a wider audience to seek social support from others [131]. It is possible that experiencing more stressors could dampen the negative relationship between audience size and distress disclosure, so a user may want to disclose distress to more people when they are highly stressed. Such sharing may help individuals obtain more informational and emotional support from others. Similarly, the positive relationship between relational closeness and distress disclosure may be weakened by several stressors, so individuals would disclose distress to weak ties in times of stress, mainly to obtain catharsis.

Furthermore, non-situational factors and individual characteristics may affect how situational factors influence disclosure behavior [89]. Gender differences may exist regarding the effects of affordances on distress disclosure. For instance, males may consider anonymity and visibility control more important in sharing negative emotions because of the stereotypes of masculinity [46]. Moreover, privacy concerns may interact with affordances in predicting

distress disclosure, such that the positive relationships between persistence, anonymity, visibility control, and distress disclosure might be weaker when privacy concerns increase.

While these speculations may seem reasonable, there is scant empirical evidence to support these interaction effects. Thus, we pose the following research question:

R2: To what extent do interactions between various situational factors and between situational and non-situational factors predict the depth of distress disclosure on social media?

2.5 Distress Disclosure and Satisfaction

What happens after people share their distress with others? Empirical evidence suggests that sharing negative emotions and feelings has social and emotional ramifications. On one hand, distress disclosure might decrease stress and anxiety through the effects of catharsis [98] and the cognitive reappraisal of the event [104]. It may also help in obtaining emotional support and instrumental assistance from others [104, 131]. Conversely and paradoxically, distress disclosure could also increase negative affect and diminish social support [28, 130] for several reasons. First, sharing negative emotions can make a person seem less attractive and amiable, thus alienating supporters [27, 30]. Second, thinking about and expressing negative thoughts or feelings can facilitate rumination, increasing the salience and memorability of the distressing event [24, 86, 94].

To explore the dilemma of distress disclosure [30], scholars have sought to determine the factors that influence the link between distress disclosure and its outcomes [7, 131]. Most notably, affordances of social media have been found to affect how people feel after they share emotions [14, 28]. In particular, sharing distress on anonymous sites may fulfill users' needs for privacy and augment social support [7, 132]. For instance, users who anonymously engaged in self-disclosure about sexual abuse on Reddit tended to receive more supportive responses than they received from making offline disclosures [7]. Content persistence may also influence disclosure satisfaction. Posting messages that are persistent and retrievable often enables the more deliberate crafting of content compared with spontaneous interactions [11, 56], which may facilitate cognitive reassessment and self-reflection similar to the effects of expressive writing, such as journaling [98]. Thus, distress disclosure on platforms that afford high content persistence might increase satisfaction. Finally, because of the need for privacy and control in distress disclosure [8], users are likely to feel more satisfied after disclosing their negative experiences and feelings on platforms that allow more visibility control of the content. Hence, we hypothesize the following:

H4: Perceived a) anonymity, b) persistence, and c) visibility control have an indirect effect on satisfaction mediated by the depth of distress disclosure.

Moreover, disclosure satisfaction may hinge on relational closeness to the audience. The social penetration theory posits that intimate disclosure between close partners can induce positive and satisfactory feelings and drive forward relational development [114]. Similarly, studies showed a positive relationship between perceived closeness to others and relationship satisfaction across friendships and romantic relationships [91]. In contrast, individuals, especially marginalized groups, are subject to cyberbullying from strangers online, which may take a toll on their mental health [45]. Therefore, it is reasonable to assume that disclosing one's distress in greater depth with relationally closer audience could enhance satisfaction and positive emotions:

H5: Relational closeness has an indirect effect on satisfaction mediated by the depth of distress disclosure.

3 METHODS

3.1 Procedure and Participants

We conducted an online survey of a nationally representative sample of American adults in May 2020. Participants were recruited via Qualtrics panels, which mainly comprised online samples from market research panels. Eligible participants were 18 years or older, fluent in English, lived in the U.S., and have used social media (including SNSs, IM, and video chat platforms) to communicate with others over the past week. Each participant was paid $1.5-2 for completing the survey, with the median response time being 9.4 minutes (M = 10.8, SD = 6.3). This study was approved by the university institutional review board (IRB) and online consent was obtained from participants.

We first conducted pilot testing of the questionnaire with 50 individuals to test and improve research design. A total of 714 respondents participated in the main study. Several strategies were used to assure data quality, including (1) limiting each account to take the survey only once; (2) removing speeders' responses, measured as less than one-half the median completion time (in this case less than 5 minutes); (3) removing survey responses of less than 70% of completion. As a result of data cleaning, 146 cases were removed, yielding 588 responses. Given our focus on distress disclosure in this study, we excluded 118 responses from respondents who had never used social media to disclose distress related to COVID-19.

The final sample consisted of 470 respondents, which represented the U.S. population in key demographic characteristics. The respondents were 53.2% female, 44.7% male, and 1.9% non-binary, with an average age of 41 (SD = 17.1), ranging from 18 to 82 years. The majority (71.1%) were White, with 14.5% African American, 6.6% Asian, 1.5% American Indians or Alaska Native, 1.3% Native Hawaiian, 3.2% had more than one race, and 1.9% were missing. About half (48.1%) of the respondents were employed and 45.9% of the participants held at least a bachelor's degree.

The online supplementary materials (OSM) of this study, including the data, questionnaire, measures, and analysis scripts, can be found at: https://osf.io/8we2s.

3.2 Measures

Platforms for distress disclosure. We asked participants to indicate if they had shared or talked about distress (e.g., stress, worry, fear, sadness) related to COVID-19 with others over the past week via a list of 17 social media platforms, including Facebook, Twitter, Instagram, Snapchat, Reddit, Tumblr, Pinterest, TikTok, YouTube, WhatsApp, WeChat, Google Hangouts, FaceTime, Skype, Zoom, Slack, and Discord. This list included eight most popular social media platforms in the U.S. [101], video chat platforms often used during COVID-19 (e.g., Zoom, Skype, FaceTime), and popular niche platforms (e.g., TikTok, Discord). If none of the listed platforms

was used, they could select "Not Applicable" or "Other" and specify other platforms they used.

Depth of distress disclosure. Participants were then instructed to identify one platform that they had most recently used for distress disclosure. They could select from the list of 17 platforms or specify another platform. For each platform selected, participants were asked to indicate how much information they shared along three dimensions: (1) details of the event, (2) personal feelings, and (3) personal thoughts. Each item was measured on a 5-point scale (1 = "Very little" to 5 = "Very much"). This measure was adapted from [28], with a Cronbach's alpha of .88 (M = 4.26, SD = 1.54).

Perceived affordances. For each platform selected, we asked participants to describe the platform or the specific feature (e.g., Facebook Messenger, Instagram Stories) they used to share or talk about their distress related to COVID-19. Participants rated how strongly they agreed or disagreed with nine statements about the platform/feature they used on a 7-point Likert scale (1 = "Strongly disagree" to 7 = "Strongly agree"). Adapted from [39, 56], these items measured the perceived persistence, anonymity, and visibility control, respectively. *Perceived persistence* was measured with three items such as "I can retrieve past messages in this platform/feature" (α = .88, M = 4.58, SD = 1.53). *Perceived anonymity* was measured with three items such as "This platform/feature can make me anonymous to the person I am communicating with" (α = .87, M = 3.68, SD = 1.60). *Perceived visibility control* was measured by three items such as "This platform/feature allows me to control who can see my communication" (α = .84, M = 4.86, SD = 1.35).

Audience size. For each platform selected, participants indicated how many people had seen or might have seen their message/post on a 7-point scale (1 = "Only one" to 7 = "More than 500"), using the measure from previous studies [131]. This measure captured either the actual or imagined audience of individuals' distress disclosure, M = 3.26, SD = 1.67.

Relational closeness. Participants reported on how close they felt to the person(s) they communicated with on a 7-point Likert scale, ranging from 1 = "Not close" to 7 = "Very close" (M = 4.89, SD = 1.66.) This measure was adapted from [121].

Disclosure satisfaction. Based on [14], we measured participants' satisfaction after disclosure with two questions: "To what extent were you satisfied with the responses you received from the person(s) you communicated with?" and "To what extent were you satisfied with the outcome of your communication?" . These questions were all measured on a 7-point scale, with 1 = "Not at all satisfied" and 7 = "Extremely satisfied" (α = .83, M = 5.29, SD = 1.35).

COVID-related stressors. Based on the literature on stressful life events and news reports related to COVID-19 [48, 69], we developed a list of 15 stressors related to COVID-19 and asked participants to check all the stressors they had experienced over the past 30 days (M = 2.93, SD = 2.38). They could also select "none of them" and/or "other" and specify any additional stressors. A complete list of stressors can be found in the Results section.

Covariates. We also included a few variables as covariates. (1) *Demographics.* As demographics and socioeconomic status can affect social media use [20, 65], the current study included gender, age, race, education, and income as covariates. (2) *Living alone or not.* Whether or not a person is living alone may also influence their disclosure behavior on social media during the pandemic, and thus was included as another control variable. (3) *General tendency of distress disclosure.* We also controlled for individuals' propensity of distress disclosure [75], measured with three items such as is "When something unpleasant happens to me, I often look for someone to talk to". These statements were rated on a 7-point Likert scale (1 = "Strongly disagree" to 7 = "Strongly agree"), with a Cronbach's alpha of .86, M = 4.45, SD = 1.45. (4) *Online privacy concerns.* Online privacy concerns are viewed as critical predictors of online disclosure [42]. Based on [89], six items were used to assess both vertical privacy concerns (e.g., "Concerned about website or app providers sharing your data with unknown third parties") and horizontal privacy concerns (e.g., "Concerned about other people finding information about you on the Internet"). Items were rated on a 7-point scale from "Strongly disagree" to "Strongly agree"), with a Cronbach's alpha of .90, M = 3.33, SD = 1.04.

3.3 Data analysis

We tested the factorial validity of all measures by running confirmatory factor analysis (CFA) with robust maximum likelihood estimation. Drawing on common goodness-of-fit (GOF) criteria by Kline (2015), we found that all main study variables showed good validity and reliability. The overall model also showed a good fit (χ^2 = 187.47, df = 94, χ^2/df = 2, p < .001, CFI = .97, TLI = .96, RMSEA = .05, SRMR = .03). We tested our hypotheses and examined the RQ using multiple regression analyses. To estimate effect sizes, the correlation coefficient r was used, where .5 signals large effects, .3 represents medium effects, and .1 describes small effects [31]. For the analyses, we used the software R (version 3.6.3) and packages such as *tidyverse, lavaan, mediation.*

4 RESULTS

4.1 Descriptive Analyses

We first presented descriptive data of the self-reported stressors and social media platforms used for distress disclosure related to COVID-19. Figure 1 showed that "Less social interactions with people" was the most commonly reported stressor (46.6% of survey respondents). Other stressors include loss of job or income, major change in recreation type and/or amount, having family/friends infected with the coronavirus, major change in eating or sleeping habits, problems getting food or household essentials, major change in work routines, etc. In terms of social media platforms used for distress disclosure, participants reported using an average of three platforms for distress disclosure. As shown in Figure 2, Facebook dominated the distress disclosure landscape, reported by 70% of respondents. This was followed by Instagram (36%), YouTube (29%), Snapchat (28%), FaceTime (25%), Twitter (24%), Zoom (20%), WhatsApp (18%), TikTok (16%), etc.

Before testing the hypotheses, we explore the correlations between the main study variables, and the results can be found in OSM.

4.2 Predicting the Depth of Distress Disclosure

RQ1 asked if specific platforms predict the depth of distress disclosure. We ran a linear regression model that included the individual platforms that participants identified as the most recent platform

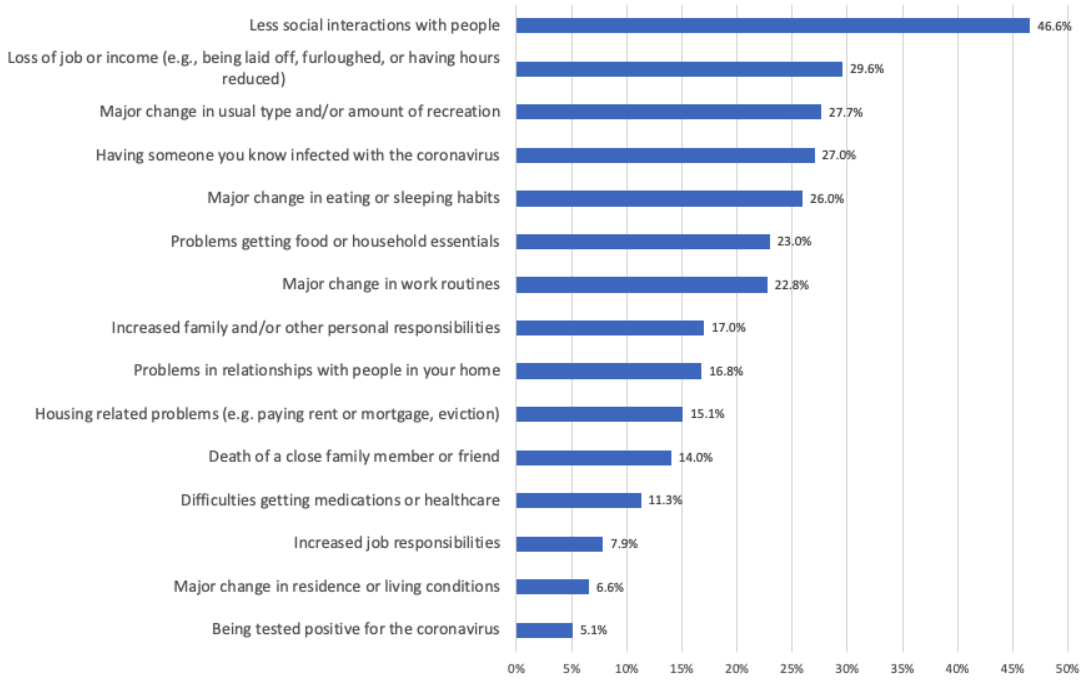


Figure 1: Frequency of self-reported stressors during COVID-19

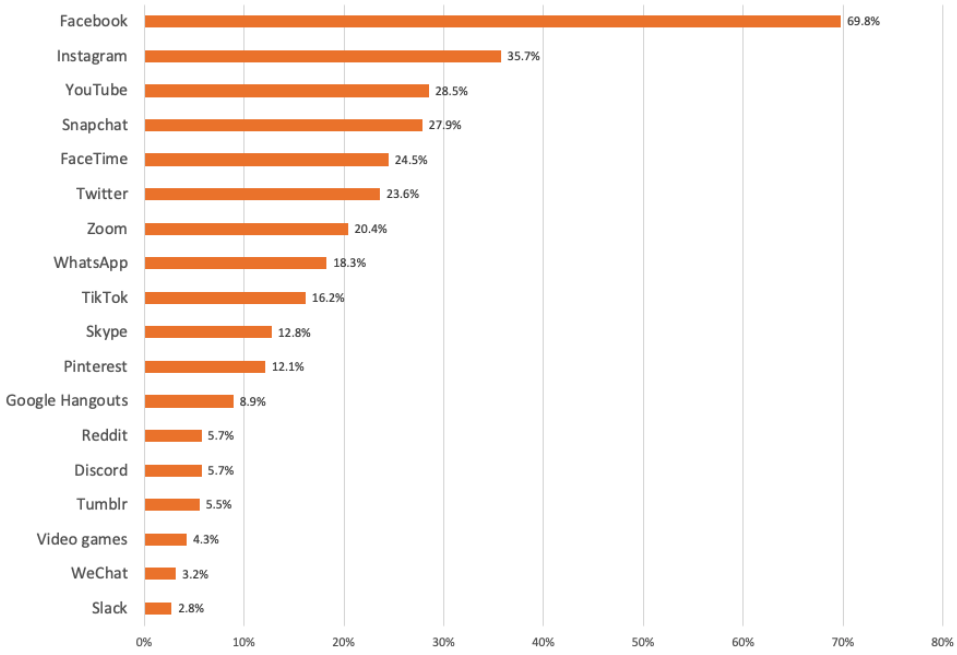


Figure 2: Frequency of self-reported social media platforms used for distress disclosure

used for distress disclosure as dummy variables, as well as other factors and covariates, to predict disclosure depth. As shown in Table 1 (Model 1), only Zoom was significantly and positively related to distress disclosure ($\beta = .10$, $SE = .05$, $p < .05$), suggesting that people disclosed more distress on Zoom. Other platforms did not significantly predict depth of distress disclosure.

H1 assumed a positive relationship between perceived affordances and distress disclosure. The results showed that the depth

Table 1: OLS regression models predicting distress disclosure and satisfaction

Predictors	Depth of Distress Disclosure			Satisfaction
	Model 1 (main effects) β [2] (SE)	Model 2 (interaction1) β (SE)	Model 3 (interaction2) β (SE)	Model 4 β (SE)
Non-situational factors [1]				
Age	-.10(.05)*	-.09(.05)	-.09(.05)	.10(.05)
Gender (Female vs. Male)	-.30(.08)*** [3]	-.29(.08)*** [3]	-.32(.08)*** [3]	.15(.09) [3]
Race (White vs. Black)	.25(.12)* [3]	.21(.12) [3]	.23(.12)* [3]	.32(.14) * [3]
Disclosure tendency	.21(.05)***	.21(.05)***	.21(.05)***	.09(.06)
Situational factors				
Personal factors				
Stressors	.05(.04)	.05(.04)	.05(.04)	-.04(.04)
Interpersonal factors				
Audience size	.06(.04)	.07(.04)	.06(.05)	-.06(.05)
Relational closeness	.17(.05)***	.20(.05)***	.17(.05)***	.34(.05)***
External factors				
Platforms [1]				
Zoom	.10(.05)*	.09(.05)	.10(.05)*	.06(.06)
Tumblr	.002(.04)	.003(.04)	.008(.04)	-.12(.04)**
Perceived Affordances				
Persistence	.24(.05)***	.24(.05)***	.24(.05)***	.13(.06)*
Anonymity	.15(.05)**	.14(.05)**	.14(.05)**	.03(.05)
Visibility	.13(.05)**	.12(.05)**	.18(.05)***	.11(.05)*
Interaction terms [1,4]				
Visibility X Closeness		.11(.04)*		
Visibility X Tendency			.12(.04)**	
Depth of disclosure				.13(.06)*
R^2	*.51*	*.52*	*.52*	*.53*
Adjusted R^2	*.46*	*.47*	*.47*	*.46*

Note: *p<0.05; **p<0.01; ***p<0.001.

[1] Only variables with significant results are presented

[2] Standardized coefficient beta (β) - continuous variables were standardized to make the effect sizes comparable.

[3] Unstandardized coefficient B – categorical variables were not standardized

[4] Only one interaction term was entered at a time in each regression

of distress disclosure was positively associated with perceived persistence (β = .24, *SE* = .05, p <.001), anonymity (β =.15, *SE* = 0.03, p <.01), and visibility control (β = .13, *SE* = .03, p <.01), lending support to H1a, H1b, and H1c. This suggested that perceived affordances of platforms might influence how much people talk about their distress, beyond the effects of platforms.

H2 proposed that disclosure depth was negatively related to audience size and positively related to relational closeness. Results in Model 1 showed that audience size was not significantly associated with distress disclosure amount (β = .06, *SE* = .04, p > .05), refuting H2a. Closeness to audience was significantly and positively associated with distress disclosure amount (β = .17, *SE* = .05, p <.001), lending support to H2b. This meant that people talked about distress in greater depth with stronger ties.

H3 predicted that the number of stressors was positively associated with depth of distress disclosure. Results showed a non-significant relationship between stressors and disclosure depth (β = .05, *SE* = .04, p > .05). This suggested that external stressors did not influence disclosure intimacy on social media.

In addition, several non-situational factors were significantly related to disclosure depth. Individuals who disclosed distress in greater depth on social media tended to be younger (β = -.10, *SE* = .05, p <.05), male (compared to female; β = -.29, *SE* = .08, p <.001), White (compared to African American; β = .26, *SE* = .12, p <.05), and had a higher propensity of distress disclosure (β = .21, *SE* = .05, p <.001). Combined, the situational and non-situational factors explained 46% of the variance in the depth of distress disclosure.

4.3 Moderation Analyses

R2 examined the interactions between situational and non-situational factors predicting distress disclosure. We first focused on interactions between same-level factors. Given that perceived affordances and relational closeness were significant predictors of disclosure, their interaction terms were added to the model. To

avoid multicollinearity issues with the interaction terms, all continuous variables were centered in the moderation analysis [1]. The results in Table 1 (Model 2) showed a significant interaction between visibility control and closeness on distress disclosure (β = .11, SE = .04, p < .05), suggesting that the positive relationship between visibility control and disclosure depth were contingent on relational closeness. Plotting the interaction terms was carried out to scrutinize the moderating effect, which confirmed interaction between visibility control and relational closeness (Figure 3). None of the other interactions were significant.

We also examined the interactions between situational and non-situational factors on distress disclosure. Since gender and disclosure tendency emerged as strong predictors, we included their interaction terms with perceived affordances in the model (Model 3). The results showed that disclosure tendency moderated the relationship between visibility control and disclosure depth (β = .12, SE = .04, p < .01), suggesting that perceived visibility control had a stronger effect on the depth of distress disclosure when individuals who were more inclined to open up to others.

4.4 Predicting Disclosure Satisfaction

H4 proposed an indirect relationship between perceived affordances and satisfaction after disclosure. We tested the significance of indirect effects using bootstrapping procedures (1,000 simulations). The results showed that satisfaction was indirectly associated with persistence (b = .03, CI [.008 to .06], p < .01), anonymity (b = .03, CI [.006 to .05], p < .01), and visibility control (b = .02, CI [.001 to .04], p<.05). Thus, H4 was supported. Additionally, Model 4 in Table 1 showed that persistence (β = .13, SE = .06, p < .05), visibility (β = .11, SE = .05, p < .05), and disclosure depth (β = .13, SE = .06, p < .05) were directly associated with satisfaction.

H5 assumed that relational closeness indirectly affected satisfaction via disclosure depth. This was supported by mediation analysis showing an indirect effect of closeness on satisfaction (b = .03, CI [.009 to .06], p < .01). Results in Table 1 (Model 4) revealed that closeness was also directly related to satisfaction (β = .31, SE = .06, p < .001). Overall, the direct and indirect link between relational closeness and disclosure satisfaction suggested that confiding in close family members and friends had positive outcomes.

4.5 Post-Hoc Analyses

Because previous research has suggested variations in user perception of social media affordances [39], we ran additional analyses to explore the factors that influenced perceived affordances. The results suggested that perceived persistence was rated higher by respondents who were White (β = .31, SE = .14, p < .05); perceived anonymity was rated higher by those who were younger (β = -.16, SE = .06, p < .01) and male (β = -.36, SE = .10, p < .001); and perceived visibility control was rated higher by respondents with higher income (β = .14, SE = .05, p < .01). These perceived affordances also differed across the 17 social media platforms. Full results can be found in OSM.

We have also explored the factors contributing to variations in the number of platforms used by respondents. The results showed that the number of social media platforms used for distress disclosure was positively associated with the number of stressors (β = .13, SE = .05, p < .01), higher education (β = .20, SE = .05, p < .001), and a higher tendency of disclosure (β = .16, SE = .05, p < .001), and was negatively related to age (β = -.26, SE = .05, p < .001). This suggested that people's disclosure ecology is predicted by a combination of situational and non-situational factors.

5 DISCUSSION

This study examined distress disclosure on social media during the COVID-19 pandemic to investigate the unique influences and interplay of situational and non-situational factors in predicting disclosure depth. Our findings indicated that perceived affordances (persistence, anonymity, and visibility control) and relational closeness with audience were the strongest predictors of distress disclosure, which affected disclosure satisfaction. In the following sections, we discuss the theoretical contributions of these findings and their implications for the design of social media to better distress disclosure.

5.1 Distress Disclosure in Social Media Ecology

HCI research on crisis informatics has long examined the role of information technology in helping people seek support and build resilience during crises [70, 87, 106, 109]. This study builds on and extends this stream of research by examining distress disclosure on social media during the COVID-19 pandemic. Given the heightened stress and disruption in offline social interactions, it was unsurprising that pandemic-related distress disclosure on social media was prevalent (80% of our survey respondents). While the disclosure of negative experiences and emotions may threaten impression management and is often subject to self-censorship on social media [28, 90], distress disclosure may have a different meaning during the pandemic. The need for social connections and support may take precedence over privacy concerns during a crisis, leading to higher levels of distress disclosure. Indeed, previous studies showed that people may loosen their privacy boundaries and use fewer privacy-protecting strategies on social media in times of stress or crisis [18, 133]. These findings indicate that social media platforms should launch timely features to support distress disclosure while preserving users' privacy, such as more context-specific emojis (e.g., wearing a mask) and reminding users about the number of users in the audience who will see their post.

Notably, our results suggest that individuals utilize multiple social media platforms (an average of three) to disclose distress, which highlights the importance of understanding the personal social media ecosystem within which people self-disclose [19, 40, 134], what we term a *social media disclosure ecology*. The results of the post-hoc analysis showed that the number of platforms people used was associated with the number of stressors they experienced. Indeed, the impact of the pandemic on people's lives varied immensely, as evidenced by the wide range of the reported number of stressors (1–15), suggesting that some people might be riding through the pandemic smoothly, while others are confronting devastating storms. These findings highlight the social embeddedness of technology [125]; that is, the use of technology is contingent on personal life circumstances and social context. While previous research suggested that the size and composition of a user's media repertoire were contingent on personal preferences and structural factors [19, 113],

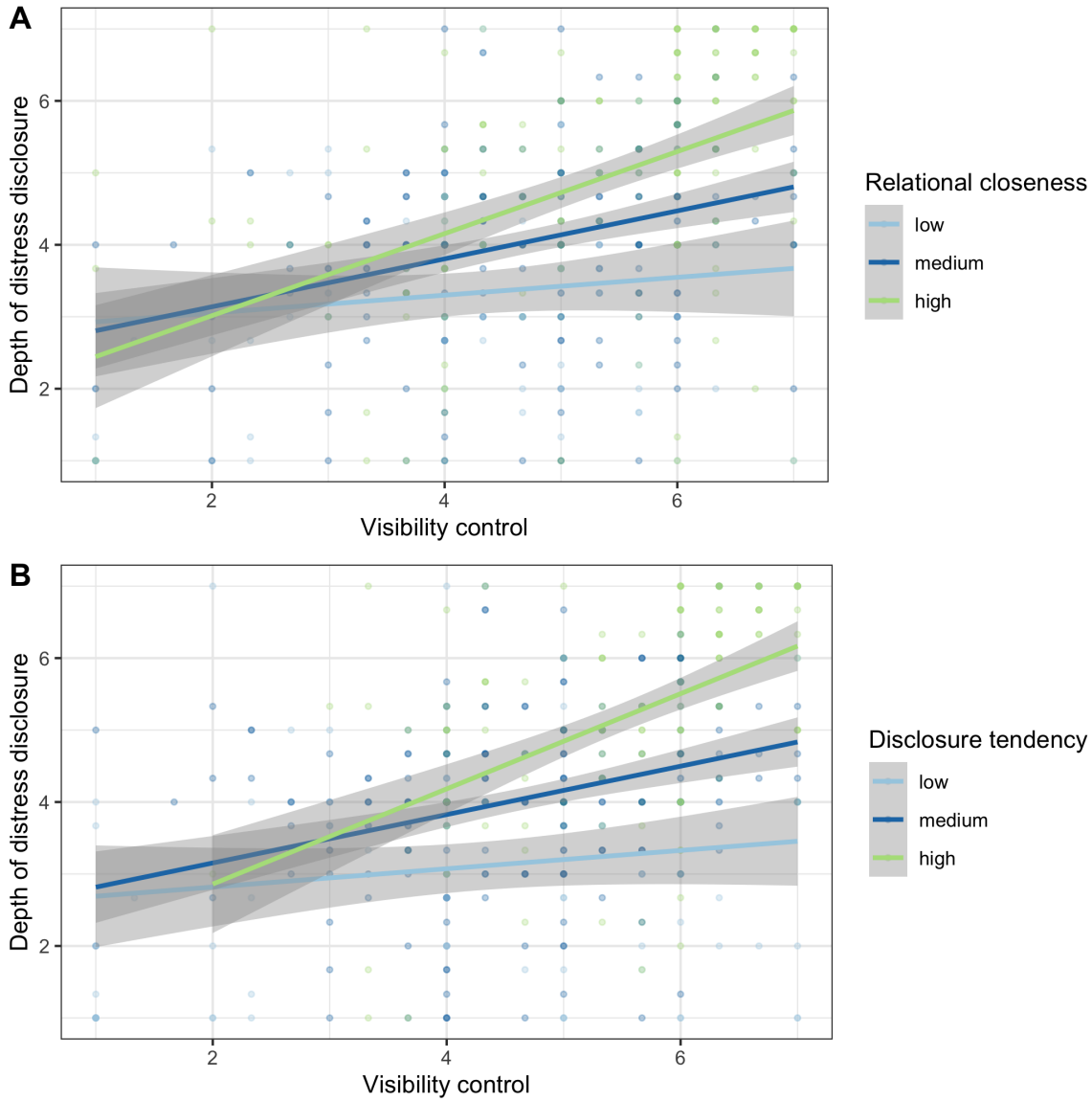


Figure 3: Interaction plots. (A) Moderating effect of relational closeness on the relationship between visibility control and the depth of distress disclosure; (B) Moderating effect of disclosure tendency on the relationship between visibility control and the depth of distress disclosure

this study extends this line of research by pinpointing the influence of situational factors on media repertoires.

It is important to note the uniqueness of pandemic-related distress disclosure. Previous research on distress disclosure has primarily focused on two types of distress. The first type concerns emotionally difficult and stigmatized experiences, such as mental illnesses and sexual abuse [6, 40, 58], which are often kept secret and are less likely to be shared publicly. When this type of distress is shared on social media, users tend to engage in selective, anonymous, and reciprocal disclosures [6, 40, 58, 132]. The second type of distress is marked by generic negative emotions, such as stress, sadness, and anger, which are likely to be more mundane and less sensitive than the first type of distress and thus shared more often with others [15, 28, 62, 131]. Such distress disclosure is often driven by the need for catharsis and support-seeking [15, 123], which, in some cases, may be impulsive [123]. This study focused on the second type of distress disclosure because pandemic-related distress could be experienced universally and therefore carry less stigma, thereby reducing the barriers to disclosing it. Thus, the findings of this study should be interpreted in its context. We discuss also how our findings corroborate or contradict previous findings on other types of distress disclosure in the following section.

5.2 The Effects of Perceived Affordances and Audience

Our findings also suggest that perceived affordances and relational closeness to audiences could directly and interactively predict the depth of distress disclosure on social media. Although previous research has provided valuable insights into individuals' decision-making processes regarding distress disclosure [6, 8, 40], the relative importance of different decision factors has remained unclear. This study extends this line of research by highlighting interrelatedness and relative strength of factors that influence distress disclosure.

Notably, perceived affordances emerged as the strongest predictor of the depth of distress disclosure in our study. The results showed that people disclosed more pandemic-related distress on platforms with more perceived anonymity, content persistence, and visibility control, beyond the effects of specific platforms. This result underscores the importance of understanding affordances in terms of user perception [33], which may vary depending on users' digital literacy, life and technology experiences, and physical and mental capability [33, 39, 66]. Our post-hoc analysis showed that indeed perceived affordances were related to the participant's age, gender, race, and income, controlling for specific platforms. This calls for attention from researchers and designers to identify discrepancies between technical features and user perception and to reflect on whether certain affordances systematically privilege some people over others [33]. For example, studies have found that older adults who are tech-savvy benefit more from online social interactions than those who have fewer Internet skills [93]. It is possible that certain affordances such as visibility control may not be perceptible to those with lower digital literacy, who may have difficulty navigating the intricate privacy settings of platforms. Recent discussions of design justice underscore that making platform affordances perceptible and accessible to all users is crucial in promoting privacy, equity, and justice [33].

In contrast to cross-platform research that suggested notable differences in user behavior across platforms [29, 105], our findings revealed a limited effect of platforms on distress disclosure. This might be because of the increasingly diversified user behavior within a platform that incorporates the wider range of affordances (e.g., anonymity, visibility, and ephemerality) [13, 50, 110], as well as the prevalence of multiple account practices [44, 63, 105]. In other words, users may exhibit multifaceted and contingent behavior on a platform, depending on the specific channel (e.g., private vs. public) and the account (e.g., anonymous vs. identified) they use. The perceived affordance approach may be especially potent in capturing the complexity of user behavior within platforms while also accounting for platform differences. Understanding higher-level attributes that transcend specific technologies may provide valuable insights into user behavior around ever-evolving technologies and generate durable theories as technologies wax and wane over time [12, 39, 51, 55, 116].

Our findings also confirm and extend the work of DeVito et al. [40], which showed that platforms and audiences often coalesce into "spaces" that fulfill individuals' particular needs. Our results revealed an interaction effect between perceived visibility control and relational closeness to audience in predicting the depth of distress disclosure. In other words, the participants disclosed their distress in more depth with stronger ties in platforms affording more visibility control. Similar to offline spaces where people chat with a few close friends privately (e.g., living room or car), virtual spaces that allow privacy and control can also prompt them to open up and disclose their innermost feelings. Hence, private messaging, one of the main social media elements [12], might be an optimal channel for intimate disclosure. Social media platforms need to make it easier for users to segment the audience, control privacy boundaries, and initiate private conversations with others.

Moreover, these findings are congruent with prior studies on sensitive and stigmatized disclosure regarding the importance of anonymity, visibility control, and audience closeness [7, 54, 132], suggesting that the need for privacy and control may cut across different types of distress disclosure at varied levels of sensitivity. The positivity bias in social media interactions still persists during a crisis, despite heightened stress and shifting norms. Although distress disclosure on social media may be more prevalent during the pandemic compared with normal times, people are likely to use similar strategies to regulate their privacy. Future research could examine how individuals' privacy practices during crises differ from those during normal times.

Recent advances in computational techniques have spurred extensive research on extracting digital trace data from platforms to model and detect distress, while generating interventions for at-risk individuals [26, 37]. These endeavors have been translated into real-world applications for suicide prevention on social media [135]. However, growing concerns have been raised regarding bias, privacy, and ethics in these computational endeavors [10, 23, 25, 127]. Our findings emphasized users' needs for privacy, anonymity, and control when they disclose distress online, which aligns with previous qualitative research on stigmatized disclosure on social media [8]. Thus, algorithmic detection and interventions based on platform data without users' explicit consent may thwart users' expectations of privacy, undermine trust in platforms, and exacerbate their distress [23, 25, 53]. For individuals who do not seek formal care due to concerns about stigma or negative experiences, mental health interventions may invoke feelings of fear, annoyance, and resistance [23]. Despite many successes in suicide prevention [124], challenges persist as algorithms are not adept at recognizing the intentions and emotional experiences underlying users' social media posts, which is compounded by methodological pitfalls and algorithmic biases [49]. Hence, extensive efforts are needed to understand the experiences and behavioral markers of individuals who are in need of professional help and, more importantly, to develop regulatory frameworks and ethical guidelines for researchers [80].

5.3 Designing to Improve Disclosure Outcomes

The findings of this study also provide insights into the outcomes of distress disclosure. Our results affirmed the positive effects of distress disclosure on satisfaction, including satisfaction with the response and the overall outcome. One potential explanation is that distress disclosure helps people unburden themselves and alleviate stress, which leads to greater satisfaction [32, 98, 111]. Distress disclosure may also help people obtain informational and emotional support from others, which fosters intimacy and trust, thereby contributing to elevated satisfaction [114]. This beneficial outcome

may be especially salient in times of crises, such as the COVID-19 pandemic. While prior work suggested that excessive distress disclosure may diminish social support and exacerbate negative emotions [27, 30, 130], our results show the opposite, probably because of the reduced judgment and stigma regarding pandemic-related distress caused by external, unpredictable, and uncontrollable forces, which is collectively shared.

Notably, we found that perceived affordances had both direct and indirect effects on disclosure satisfaction. While previous research has examined the psychological, emotional, and relational outcomes of distress disclosure on social media [7, 14, 28, 132], there is little evidence on the effects of perceived affordances on disclosure outcomes. Our results suggest that distress disclosure on platforms with higher perceived persistence, anonymity, and visibility control contributes to greater satisfaction. The positive effects of perceived content persistence on satisfaction might be linked to the process of crafting retrievable messages, which can foster cognitive reassessment [98] and allow people to review and reflect on their thoughts, contributing to emotional relief and self-reflection [71]. Our results also corroborate prior work that indicated the importance of anonymity in the public disclosure of stigmatized experiences, such as mental illness and sexual abuse, on social media [7, 132]. Finally, the positive effects of visibility control on satisfaction strongly indicate the importance of privacy in distress disclosure [8].

In light of the effects of these affordances on satisfaction, design features that allow for anonymity, visibility control, and content persistence would facilitate positive outcomes of distress disclosure. To increase users' perceived visibility control of their social media content, platforms need to increase the transparency and usability of their privacy settings, making it intuitive and effortless for users to segment audiences and adjust visibility options. Because individuals' perceived affordances may depend on their digital literacy, designers need to consider people who are less tech-savvy in designing their privacy features. Moreover, given the increasing popularity of maintaining multiple accounts on one platform, such as Finsta (a.k.a. fake Instagram), for anonymous posting [41, 76], platforms need to ensure that the option of anonymity is clearly identified. An example of the intrusion on user anonymity is the algorithmic suggestion of a user's Finsta account to friends. It is important for platforms to allow users to protect their accounts from being suggested to others. Finally, in light of the positive outcomes of content persistence, ephemeral platforms such as Snapchat may consider ways of allowing the posting of persistent messages that may enhance the positive outcomes of distress disclosure.

5.4 Limitations and Future Work

This study has several limitations. First, because our data were self-reported, they may be subject to recall bias and social desirability bias. Thus, the findings should be interpreted as individuals' perceived distress disclosure rather than their actual behavior. Our choice of self-report data was guided by two reasons. First, the collection of users' behavioral data across multiple platforms is extremely difficult and may raise privacy concerns. Second, prior research suggested that individuals often make implicit and indirect disclosures on social media, especially regarding negative feelings [8]. This type of disclosure is easier to be captured by individuals than by algorithms. Future research could consider employing a combination of app logging, experience sampling methods, and surveys to passively track user behavior while also eliciting users' self-reports in real time, which could largely reduce recall bias [73, 89].

Second, given the cross-sectional nature of this study, we gained limited understanding of the evolution and adaptation of personal disclosure ecology over time, especially during crises. The long-term effects of distress disclosure are also unknown. Therefore, longitudinal studies are needed to examine the variability in social media use, disclosure behaviors, and disclosure outcomes [79, 105, 115]. For example, it would be helpful to employ experience sampling methods to capture users' thoughts, feelings, and behaviors on multiple occasions over a certain period of time [21, 73, 89]. Qualitative research is needed to investigate how people create, maintain, and adapt their social media ecology in times of stress.

Third, this study focused on three types of affordances that were the most relevant in the context of distress disclosure. Future research could explore other affordances that may also play a role in online disclosure. For example, the network association affordance that allows users to create, view, and traverse social connections among users on a site [39, 56, 116] might also be related to distress disclosure, as people seek to obtain support and resources during crises. Another key social media affordance is visibility, which refers to the extent to which an individual's activity, presence, and communication are visible to others [116, 117]. We did not include this affordance in our analyses and focused instead on visibility control because people often use a variety of strategies to limit the visibility of their content despite the ability to make it public [105]. Thus, we believed visibility control might be a more predictor of disclosure depth than visibility. However, future research could compare and contrast the effects of both visibility and visibility control on online disclosure.

Finally, this study focused on distress disclosure in the US context, which might limit the generalizability of our findings to other populations. While distress has increased worldwide during COVID-19, the US context is different from contexts in Europe or Asia, specifically because of the intersection of pandemic-related distress and other stressors, such as racial violence, social justice protests, and the presidential election. Thus, it is impossible to separate COVID-related distress from other types of distress. Cross-cultural studies are needed to examine if the findings of this study can be applied to other contexts.

6 CONCLUSION

Despite a plethora of research on the factors that influence decisions and behavior of online self-disclosure, less is known about the relative importance and interplay of these factors. In particular, as platforms incorporate a wider range of affordances that enable more diverse user behaviors and audience segmentation, it is important to untangle the effects of platforms, affordances, and audiences. In this study, we examined how a variety of factors directly and interactively predicted distress disclosure on social media and disclosure satisfaction. The results suggested that perceived affordances and relational closeness were the strongest predictors of the depth of

distress disclosure, which in turn affected users' satisfaction after disclosure. This study contributes to social computing research and provides implications for the design of social media platforms to better support people in distress.

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